

MINUTES OF MEETING (MOM)

Team 8

3D Printing Cost Estimation Platform

Meeting Details

Project: 3D Printing Cost Estimation Platform
Week: Week 10
Date: Friday, April 10, 2026
Time: 4:00 PM IST
Location: CIE, Ground Floor
Client: Anil Kumar Upadhyay, Five Fingers Innovative Solutions
Client Email: aku1974india@gmail.com
Client Phone: 9949497560

Team Members Present

- Abhiroop Pullepu
- Mehrish Khan
- Ronith Menneni
- Sai Hasini
- Ruschita Guddeti

Notes

- Meeting focused on comprehensive test suite implementation, R2 feature completion, documentation updates, and preparation for final release.

1 Test Suite Implementation

1.1 Comprehensive Test Coverage

- Implemented 30 test cases covering all use cases from UC-01 through UC-15.
- Test cases organized by use case with clear naming convention (e.g., `test_uc01_tc01_login_valid`).
- All tests use in-memory SQLite database for isolation and mock external services (file system, email, Cura slicing service).
- Test runner scripts created: `run_all_tests.sh` (shell) and `run_all_tests.py` (Python).

1.2 Test Results

- **Total Tests:** 35 (30 test plan + 5 existing)
- **Passed:** 35
- **Failed:** 0
- **Skipped:** 0

1.3 Test Categories

- **UC-01 (Authentication):** 3 tests - Login validation, unauthorized access, user profile
- **UC-02 (Upload):** 3 tests - Valid STL upload, invalid file type, oversized file
- **UC-03 (3D Viewer):** 2 tests - Model rendering, large file performance
- **UC-04 (Slicing Parameters):** 2 tests - Valid parameters, missing material
- **UC-05 (Slicing Process):** 3 tests - Success, invalid geometry, service unavailable
- **UC-06 (Cost Formula):** 2 tests - Formula verification, known values calculation
- **UC-07 (Cost Display):** 1 test - Cost estimate breakdown
- **UC-08 (Error Handling):** 1 test - Slicing error display
- **UC-09 (Admin Logs):** 1 test - Failure logs dashboard
- **UC-11 (Material Config):** 3 tests - Edit material, add material, invalid cost
- **UC-12 (Machine Rate):** 1 test - Update machine time rate
- **UC-13 (Overhead Factors):** 1 test - Update waste and failure factors
- **UC-14 (Validation):** 1 test - Oversized model validation
- **UC-15 (Job History):** 1 test - Job history display
- **Integration:** 1 test - Auth workflow
- **Performance:** 3 tests - Upload speed, slicing speed, API response time
- **Security:** 1 test - Unauthorized access

2 R2 Feature Completion

2.1 Backend Improvements

- Added TESTING environment variable support for test isolation.
- Conditional table creation to avoid PostgreSQL connection during tests.
- Enhanced error handling with specific error messages.
- Improved admin failure logs dashboard with working filters.
- Material cost validation (positive numbers only).

2.2 Documentation Updates

- Updated README.md with R2 release information, comprehensive feature documentation.
- Added cost calculation formula with example.
- Added detailed project structure with test files.
- Added test suite documentation with coverage breakdown.
- Added environment variables documentation.
- Added troubleshooting section.
- Updated TestPlan.xls and created TestPlan.csv for easy reference.

2.3 Test Plan Documentation

- Test plan includes pre-conditions, step-by-step procedures, expected outcomes.
- R1/R2 implementation status tracking for each test case.
- All 30 test cases aligned with updated TestPlan.xls.

3 Code Quality and Best Practices

3.1 Test Isolation

- Pytest fixtures for database, client, users, tokens, materials, and admin config.
- Mocking of external dependencies (file operations, email sending, Cura service).
- Environment variables (LOG_DIR, TESTING) for test configuration.

3.2 Code Organization

- Test files organized by use case (UC-01 through UC-15).
- Integration, performance, and security tests in separate files.
- Conftest.py for shared fixtures and test configuration.

4 Client Discussion and Next Steps

4.1 Client Feedback

- Client reviewed the comprehensive test suite and expressed satisfaction with coverage.
- Client appreciated the detailed documentation and troubleshooting guide.
- Client requested final release be scheduled for April 17, 2026.

4.2 Final Release Planning

- Final release (R2) scheduled for April 17, 2026.
- All 30 test cases passing with 0 failures.
- Documentation complete and updated.
- Handover documentation to be prepared.

5 Action Items Summary

Owner	Task
Abhiroop Pullepu	Prepare final handover documentation including deployment guide and user manual.
Mehrish Khan	Verify all test cases pass in production-like environment.
Ronith Menneni	Finalize README and documentation for R2 release.
Sai Hasini	Create presentation for final client demo on April 17.
Ruschita Guddeti	Prepare project summary and deliverables checklist.
Client (Anil Kumar)	Schedule final review meeting for April 17, 2026.

Minutes prepared by: Team 8
Date: April 10, 2026